3	**(a)**	**(i)**	velocity = rate of <u>change</u> of displacement OR displacement <u>change</u> / time (taken)	A1	[1]
		(ii)	acceleration = rate of <u>change</u> of velocity OR <u>change</u> in velocity / time (taken)	A1	[1]
	(b)	**(i)**	initial constant velocity as straight line / gradient constant	B1	
			middle section deceleration/ speed / velocity decreases / slowing down as gradient decreases	B1	
			last section lower velocity (than at start) as gradient (constant and) smaller	B1	[3]
			[special case: all three stages correct descriptions but no reasons 1/3]		
		(ii)	velocity = 45 / 1.5 = 30 $\mathrm{m\,s^{-1}}$	A1	[1]
		(iii)	velocity at 4.0 s is (122 – 98) / 2.0 = 12 ($\mathrm{m\,s^{-1}}$) (allow 12 to 13)	B1	
			acceleration = (12 – 30) / 2.5 = –7.2 $\mathrm{m\,s^{-2}}$ (if answer not this value then comment needed to explain why, e.g. difficulty in drawing tangent)	A1	[2]
		(iv)	$F = ma$	C1	
			= (–)1500 × 7.2 = (–)11000 (10800) N	A1	[2]